THE ASCENDING CHAIN CONDITION FOR PRINCIPAL IDEALS FORCES FACTORIZATIONS INTO IRREDUCIBLES

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- An element that is neither irreducible nor a unit factors into two proper factors. Used at proof lines 29615-29619. Produces the next element of the descending divisibility sequence from a failure of irreducibility.
- The ascending chain condition for principal ideals. Contradicted at the end of the counted proof (lines 29624-29626). Forbids the strictly increasing chain of principal ideals constructed from the descending sequence.

Lemma 0.1. *Let R be a domain. Consider the following conditions:*

(1) *The ring R satisfies the ascending chain condition for principal ideals.*
(2) *Every nonzero, nonunit element $a \in R$ has a factorization $a = b_1 \ldots b_k$ with each b_i an irreducible element of R.*

Then (1) implies (2).

Proof. Let x be a nonzero element, not a unit, which does not have a factorization into irreducibles. Set $x_1 = x$. We can write $x = yz$ where neither y nor z is irreducible or a unit. Then either y does not have a factorization into irreducibles, in which case we set $x_2 = y$, or z does not have a factorization into irreducibles, in which case we set $x_2 = z$. Continuing in this fashion we find a sequence

$$\ldots | x_3 | x_2 | x_1$$

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of elements of R with x_n/x_{n+1} not a unit. This gives a strictly increasing sequence of principal ideals $(x_1) \subset (x_2) \subset (x_3) \subset \ldots$ thereby finishing the proof. $\square$